VALIDATION PLAN
OPERATIONAL QUALIFICATION
SPC MODULE

JR Validated Environment — SPC Module v1.0

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Version History

Version	Date	Author	Description
1.0	2026-03-18	Joep Rous	Initial release. Covers all 5 SPC scripts in module v1.0.

Approval Signatures

Role	Name	Signature	Date
Author			
Reviewer			
Approver (QA)			

1. Purpose

This document defines the Validation Plan for the Operational Qualification (OQ) of the Statistical Process Control (SPC) module included in version 1.0 of the JR Validated Environment. It specifies the user requirements, test cases, acceptance criteria, and traceability between requirements and tests needed to demonstrate that each SPC script performs correctly for its intended use.

Execution of the tests defined in this plan, together with satisfactory results and documented evidence, constitutes the OQ for the SPC module.

2. Scope

2.1 In Scope

- All 5 R scripts in the repos/spc/R/ directory: jrc_spc_imr, jrc_spc_xbar_r, jrc_spc_xbar_s, jrc_spc_p, jrc_spc_c
- Correct computation of control limits and all 8 Western Electric rule violations for valid inputs
- Correct rejection of invalid inputs with informative error messages
- Correct file I/O behaviour (CSV input, PNG output to ~/Downloads/)
- Verification that all R scripts enforce execution within the jrrun validated environment by checking RENV_PATHS_ROOT at startup
- Balanced-design validation for subgrouped charts (X-bar/R, X-bar/S)
- Subgroup size constraint enforcement (X-bar/R: n = 2–10; X-bar/S: n ≥ 2)
- Defectives-exceed-n validation for P-chart

2.2 Out of Scope

- Infrastructure scripts in bin/ and admin/ (covered by IQ; see JR-VP-001)
- Performance qualification (PQ) — production process monitoring studies are conducted separately under individual process control protocols
- User acceptance testing (UAT)
- Validation of R language interpreter and third-party packages (ggplot2, grid) — package integrity is confirmed at installation via SHA256 (JR-IQ-001)
- Statistical validation of the underlying SPC methods — method references are cited in Section 3

3. Regulatory Basis and References

3.1 Regulations

Reference	Title
21 CFR Part 820, §820.70(i)	Quality System Regulation — Automated Data Processing
21 CFR Part 11	Electronic Records; Electronic Signatures

3.2 FDA Guidance

Reference	Title	Date
FDA GPSV	General Principles of Software Validation; Final Guidance for Industry and FDA Staff	January 2002

FDA CSA	Computer Software Assurance for Production and Quality System Software; Draft Guidance	September 2022

3.3 Standards and Methods

Reference	Title
AIAG SPC, 2nd Ed.	Statistical Process Control Reference Manual, Automotive Industry Action Group, 2005
Montgomery (2020)	Montgomery DC. Introduction to Statistical Quality Control, 8th Ed. Wiley.
Western Electric (1956)	Statistical Quality Control Handbook. Western Electric Company, Indianapolis.
ISO 7870-2:2023	Control charts — Part 2: Shewhart control charts
ISO 7870-3:2012	Control charts — Part 3: Acceptance control charts
ISO 13485:2016	Medical devices — Quality management systems — Requirements for regulatory purposes

3.4 Internal Documents

Document Number	Title
JR-VP-001	Validation Plan — Installation Qualification, JR Validated Environment v1.0.0
JR-IQ-001	IQ Execution Evidence (docs/IQ_validation_20260311_205146.txt)
JR-VP-002	Validation Plan — OQ, Community Script Suite v1.1.0

4. User Requirements

The following user requirements define the functional behaviour that the SPC module must satisfy. Each requirement is traced to one or more test cases in Section 6.

UR	Script(s)	Requirement
UR-SPC-001	jrc_spc_imr	Compute X-bar, MR-bar, UCL and LCL for both the Individuals chart and the Moving Range chart from a valid id/value CSV. Report all values in terminal output.
UR-SPC-002	jrc_spc_imr	Apply all 8 Western Electric rules to the Individuals chart and Rule 1 only to the Moving Range chart. List each out-of-control observation by ID, value, and rule(s) fired.
UR-SPC-003	jrc_spc_imr	Accept optional --ucl and --lcl arguments. When provided, use user-specified limits in place of computed limits and label them as user-specified in output.
UR-SPC-004	jrc_spc_imr	Output a clear IN CONTROL or OUT OF CONTROL verdict based on WE rule results.
UR-SPC-005	jrc_spc_imr	Save a two-panel PNG (Individuals chart + MR chart) to ~/Downloads/ with a datetime prefix.
UR-SPC-006	jrc_spc_imr	Exit non-zero with an informative error message when: no arguments are supplied; the input file is not found; a required column is missing; fewer than 2 observations are present.
UR-SPC-007	jrc_spc_xbar_r	Compute grand mean, R-bar, and X-bar/R control limits using tabulated A2, D3, D4 constants for n = 2–10 from a valid subgroup/value long-format CSV.
UR-SPC-008	jrc_spc_xbar_r	Apply all 8 Western Electric rules to the X-bar chart and Rule 1 only to the R chart. List each out-of-control subgroup by label, value, and rule(s) fired.
UR-SPC-009	jrc_spc_xbar_r	Accept optional --ucl and --lcl arguments for user-specified X-bar limits.
UR-SPC-010	jrc_spc_xbar_r	Output a clear IN CONTROL or OUT OF CONTROL verdict.
UR-SPC-011	jrc_spc_xbar_r	Save a two-panel PNG (X-bar chart + R chart) to ~/Downloads/ with a datetime prefix.
UR-SPC-012	jrc_spc_xbar_r	Reject unbalanced subgroups with a non-zero exit and informative message. Reject subgroup size n > 10 with a non-zero exit and suggestion to use X-bar/S.
UR-SPC-013	jrc_spc_xbar_s	Compute grand mean, S-bar, and X-bar/S control limits using the analytical c4(n) formula (gamma function) for any n ≥ 2.
UR-SPC-014	jrc_spc_xbar_s	Apply all 8 Western Electric rules to the X-bar chart and Rule 1 only to the S chart. List each out-of-control subgroup by label, value, and rule(s) fired.
UR-SPC-015	jrc_spc_xbar_s	Accept optional --ucl and --lcl arguments for user-specified X-bar limits.
UR-SPC-016	jrc_spc_xbar_s	Output a clear IN CONTROL or OUT OF CONTROL verdict.
UR-SPC-017	jrc_spc_xbar_s	Save a two-panel PNG (X-bar chart + S chart) to ~/Downloads/ with a datetime prefix.
UR-SPC-018	jrc_spc_xbar_s	Reject unbalanced subgroups with a non-zero exit and informative message.
UR-SPC-019	jrc_spc_p	Compute p-bar and per-subgroup UCL/LCL from a subgroup/n/defectives CSV. Apply all 8 Western Electric rules to standardised z-values. List violations.
UR-SPC-020	jrc_spc_p	Output a clear IN CONTROL or OUT OF CONTROL verdict.
UR-SPC-021	jrc_spc_p	Reject any row where defectives > n with a non-zero exit and informative message.
UR-SPC-022	jrc_spc_p	Save a PNG (P-chart with step-line variable control limits) to ~/Downloads/.
UR-SPC-023	jrc_spc_c	Compute c-bar, sigma = sqrt(c-bar), UCL = c-bar + 3*sigma, LCL = max(0, c-bar - 3*sigma) from a subgroup/defects CSV. Apply all 8 Western Electric rules. List violations.
UR-SPC-024	jrc_spc_c	Output a clear IN CONTROL or OUT OF CONTROL verdict and save a PNG with sigma zone lines to ~/Downloads/.
UR-SPC-025	All scripts	Each script must check for the RENV_PATHS_ROOT environment variable at startup and exit non-zero with an informative message if it is absent, preventing execution outside the jrrun validated environment.
UR-SPC-026	All scripts	Calling any script with no arguments must exit non-zero and print a usage

		message showing the expected syntax.
UR-SPC-027	All scripts	Supplying a non-existent CSV path must exit non-zero with a message indicating that the file was not found.

5. Test Environment Requirements

The following environment must be in place before executing the OQ tests.

Item	Requirement
Operating system	macOS (Apple Silicon or Intel)
R version	4.5.x (as specified in admin/r_version.txt)
Python version	3.11.x (as specified in repos/spc/oq/python_version.txt)
pytest version	8.3.4 (as specified in repos/spc/oq/requirements.txt)
OQ virtual environment	~/.venvs/<PROJECT_ID>_oq/ (created by admin_msa_oq or admin_spc_oq)
jrrun	bin/jrrun present and executable; project integrity check passing
RENV_PATHS_ROOT	Set by jrrun at execution time; must NOT be set in the test shell
Test data	repos/spc/oq/data/ — 17 synthetic CSV files committed to the repo
Test runner	repos/spc/admin_spc_oq

6. Test Cases

Each test case is implemented as a pytest function in repos/spc/oq/. Tests are executed via repos/spc/admin_spc_oq, which activates the OQ virtual environment, runs pytest with -v, and captures the output to a timestamped evidence file.

6.1 jrc_spc_imr

TC-SPC-IMR-001 — Stable dataset: exit 0, key sections present

Tests: UR-SPC-001, UR-SPC-004
Command: `jrc_spc_imr oq/data/imr_stable.csv`
Rationale: Verifies the script completes successfully for a well-behaved dataset.
Pass criterion: Exit code 0; output contains control limit section, process stability section, and a verdict line.

TC-SPC-IMR-002 — Stable dataset: IN CONTROL verdict

Tests: UR-SPC-002, UR-SPC-004
Command: `jrc_spc_imr oq/data/imr_stable.csv`
Rationale: imr_stable.csv is designed so no WE rule fires.
Pass criterion: Exit code 0; 'IN CONTROL' or 'STABLE' present in output.

TC-SPC-IMR-003 — OOC dataset: exit 0, OUT OF CONTROL in output

Tests: UR-SPC-002, UR-SPC-004
Command: `jrc_spc_imr oq/data/imr_ooc.csv`
Rationale: imr_ooc.csv contains observation 13 = 11.50, clearly beyond 3σ (UCL ≈ 11.27).
Pass criterion: Exit code 0; 'OUT OF CONTROL' or 'SIGNALS' present in output.

TC-SPC-IMR-004 — OOC dataset: observation 13 flagged as Rule 1 violation

Tests: UR-SPC-002
Command: `jrc_spc_imr oq/data/imr_ooc.csv`
Rationale: Rule 1 (beyond 3σ) must fire for observation 13.
Pass criterion: Output contains '13' and one of '[1]', 'Rule 1', or 'beyond 3' (case-insensitive).

TC-SPC-IMR-005 — User-specified limits accepted

Tests: UR-SPC-003
Command: `jrc_spc_imr oq/data/imr_stable.csv --ucl 11.0 --lcl 9.0`
Rationale: Confirms --ucl / --lcl flags are parsed and reported.
Pass criterion: Exit code 0; output contains '11.0' or '11.00' or 'user' or 'specified'.

TC-SPC-IMR-006 — PNG written to ~/Downloads/

Tests: UR-SPC-005
Command: `jrc_spc_imr oq/data/imr_stable.csv`
Rationale: A PNG file must appear in ~/Downloads/ with mtime ≥ test start time.
Pass criterion: File matching *_jrc_spc_imr.png exists in ~/Downloads/ with recent mtime.

TC-SPC-IMR-007 — No arguments: non-zero exit, usage message

Tests: UR-SPC-006
Command: `jrc_spc_imr`
Pass criterion: Exit code ≠ 0; 'Usage' or 'usage' present in output.

TC-SPC-IMR-008 — File not found: non-zero exit

Tests: UR-SPC-006, UR-SPC-027
Command: `jrc_spc_imr /tmp/no_such_file_xyz.csv`
Pass criterion: Exit code ≠ 0; 'not found' or the filename present in output.

TC-SPC-IMR-009 — Missing column: non-zero exit, column named

Tests: UR-SPC-006
Command: `jrc_spc_imr oq/data/imr_missing_col.csv`
Rationale: imr_missing_col.csv lacks the 'value' column.
Pass criterion: Exit code ≠ 0; 'value' present in output.

TC-SPC-IMR-010 — Too few observations: non-zero exit

Tests: UR-SPC-006
Command: `jrc_spc_imr oq/data/imr_one_obs.csv`
Rationale: Moving range requires at least 2 observations.
Pass criterion: Exit code ≠ 0.

TC-SPC-IMR-011 — Bypass protection: direct Rscript call fails

Tests: UR-SPC-025
Command: `Rscript repos/spc/R/jrc_spc_imr.R oq/data/imr_stable.csv (RENV_PATHS_ROOT unset)`
Rationale: Script must detect absent RENV_PATHS_ROOT and refuse to run.
Pass criterion: Exit code ≠ 0; 'RENV_PATHS_ROOT' present in stderr/stdout.

6.2 jrc_spc_xbar_r

TC-SPC-XBR-001 — Stable dataset: exit 0, key sections present

Tests: UR-SPC-007, UR-SPC-010
Command: `jrc_spc_xbar_r oq/data/xbar_r_stable.csv`
Rationale: Verifies the script completes successfully for a well-behaved dataset.
Pass criterion: Exit code 0; output contains X-bar/grand mean section, R-bar section, and verdict.

TC-SPC-XBR-002 — Stable dataset: IN CONTROL verdict

Tests: UR-SPC-008, UR-SPC-010
Command: `jrc_spc_xbar_r oq/data/xbar_r_stable.csv`
Rationale: xbar_r_stable.csv is designed so no WE rule fires.
Pass criterion: Exit code 0; 'IN CONTROL' or 'STABLE' present in output.

TC-SPC-XBR-003 — OOC dataset: exit 0, OUT OF CONTROL in output

Tests: UR-SPC-008, UR-SPC-010
Command: `jrc_spc_xbar_r oq/data/xbar_r_ooc.csv`
Rationale: xbar_r_ooc.csv has subgroup sg09 with mean ≈ 53, far above UCL ≈ 50.92.
Pass criterion: Exit code 0; 'OUT OF CONTROL' or 'SIGNALS' present in output.

TC-SPC-XBR-004 — OOC dataset: subgroup sg09 flagged

Tests: UR-SPC-008
Command: `jrc_spc_xbar_r oq/data/xbar_r_ooc.csv`
Pass criterion: Output contains 'sg09'.

TC-SPC-XBR-005 — User-specified limits accepted

Tests: UR-SPC-009
Command: `jrc_spc_xbar_r oq/data/xbar_r_stable.csv --ucl 52.0 --lcl 48.0`
Pass criterion: Exit code 0; output contains '52.0' or '52.00' or 'user' or 'specified'.

TC-SPC-XBR-006 — PNG written to ~/Downloads/

Tests: UR-SPC-011
Command: `jrc_spc_xbar_r oq/data/xbar_r_stable.csv`
Pass criterion: File matching *_jrc_spc_xbar_r.png exists in ~/Downloads/ with recent mtime.

TC-SPC-XBR-007 — No arguments: non-zero exit, usage message

Tests: UR-SPC-026
Command: `jrc_spc_xbar_r`
Pass criterion: Exit code ≠ 0; 'Usage' or 'usage' present in output.

TC-SPC-XBR-008 — File not found: non-zero exit

Tests: UR-SPC-027
Command: `jrc_spc_xbar_r /tmp/no_such_file_xyz.csv`
Pass criterion: Exit code ≠ 0; 'not found' or filename present in output.

TC-SPC-XBR-009 — Missing column: non-zero exit, column named

Tests: UR-SPC-007
Command: `jrc_spc_xbar_r oq/data/xbar_r_missing_col.csv`
Pass criterion: Exit code ≠ 0; 'value' present in output.

TC-SPC-XBR-010 — Unbalanced subgroups: non-zero exit

Tests: UR-SPC-012
Command: `jrc_spc_xbar_r oq/data/xbar_r_unbalanced.csv`
Rationale: xbar_r_unbalanced.csv has subgroups of different sizes.
Pass criterion: Exit code ≠ 0; one of 'unbalanced', 'equal', or 'same size' present in output.

TC-SPC-XBR-011 — Subgroup size n > 10: non-zero exit

Tests: UR-SPC-012
Command: `jrc_spc_xbar_r oq/data/xbar_r_n_too_large.csv`
Rationale: xbar_r_n_too_large.csv has n = 11; tabulated constants only cover n = 2–10.
Pass criterion: Exit code ≠ 0; one of '10', 'xbar_s', or 'large' present in output.

TC-SPC-XBR-012 — Bypass protection: direct Rscript call fails

Tests: UR-SPC-025
Command: `Rscript repos/spc/R/jrc_spc_xbar_r.R oq/data/xbar_r_stable.csv (RENV_PATHS_ROOT unset)`
Pass criterion: Exit code ≠ 0; 'RENV_PATHS_ROOT' present in output.

6.3 jrc_spc_xbar_s

TC-SPC-XBS-001 — Stable dataset: exit 0, key sections present

Tests: UR-SPC-013, UR-SPC-016
Command: `jrc_spc_xbar_s oq/data/xbar_s_stable.csv`
Rationale: 20 subgroups × n = 8; no WE rule should fire.
Pass criterion: Exit code 0; output contains X-bar section, S-bar/Std Dev section, and verdict.

TC-SPC-XBS-002 — Stable dataset: IN CONTROL verdict

Tests: UR-SPC-014, UR-SPC-016
Command: `jrc_spc_xbar_s oq/data/xbar_s_stable.csv`
Pass criterion: Exit code 0; 'IN CONTROL' or 'STABLE' present in output.

TC-SPC-XBS-003 — OOC dataset: exit 0, OUT OF CONTROL in output

Tests: UR-SPC-014, UR-SPC-016
Command: `jrc_spc_xbar_s oq/data/xbar_s_ooc.csv`
Rationale: xbar_s_ooc.csv has subgroup sg09 with mean ≈ 102, far above UCL ≈ 100.55.
Pass criterion: Exit code 0; 'OUT OF CONTROL' or 'SIGNALS' present in output.

TC-SPC-XBS-004 — OOC dataset: subgroup sg09 flagged

Tests: UR-SPC-014
Command: `jrc_spc_xbar_s oq/data/xbar_s_ooc.csv`

Pass criterion: Output contains 'sg09'.

TC-SPC-XBS-005 — User-specified limits accepted

Tests: UR-SPC-015
Command: `jrc_spc_xbar_s oq/data/xbar_s_stable.csv --ucl 102.0 --lcl 98.0`
Pass criterion: Exit code 0; output contains '102.0' or '102.00' or 'user' or 'specified'.

TC-SPC-XBS-006 — PNG written to ~/Downloads/

Tests: UR-SPC-017
Command: `jrc_spc_xbar_s oq/data/xbar_s_stable.csv`
Pass criterion: File matching *_jrc_spc_xbar_s.png exists in ~/Downloads/ with recent mtime.

TC-SPC-XBS-007 — No arguments: non-zero exit, usage message

Tests: UR-SPC-026
Command: `jrc_spc_xbar_s`
Pass criterion: Exit code ≠ 0; 'Usage' or 'usage' present in output.

TC-SPC-XBS-008 — File not found: non-zero exit

Tests: UR-SPC-027
Command: `jrc_spc_xbar_s /tmp/no_such_file_xyz.csv`
Pass criterion: Exit code ≠ 0; 'not found' or filename present in output.

TC-SPC-XBS-009 — Missing column: non-zero exit, column named

Tests: UR-SPC-013
Command: `jrc_spc_xbar_s oq/data/xbar_s_missing_col.csv`
Pass criterion: Exit code ≠ 0; 'value' present in output.

TC-SPC-XBS-010 — Unbalanced subgroups: non-zero exit

Tests: UR-SPC-018
Command: `jrc_spc_xbar_s oq/data/xbar_r_unbalanced.csv`
Rationale: Reuses the xbar_r unbalanced dataset (same column structure).
Pass criterion: Exit code ≠ 0; one of 'unbalanced', 'equal', or 'same size' present in output.

TC-SPC-XBS-011 — Bypass protection: direct Rscript call fails

Tests: UR-SPC-025
Command: `Rscript repos/spc/R/jrc_spc_xbar_s.R oq/data/xbar_s_stable.csv (RENV_PATHS_ROOT unset)`
Pass criterion: Exit code ≠ 0; 'RENV_PATHS_ROOT' present in output.

6.4 jrc_spc_p

TC-SPC-P-001 — Stable dataset: exit 0, key sections present

Tests: UR-SPC-019, UR-SPC-020
Command: `jrc_spc_p oq/data/p_stable.csv`
Rationale: 25 subgroups, n = 100 each, defectives vary 1–8; no WE rule should fire.
Pass criterion: Exit code 0; output contains p-bar section, UCL section, and verdict.

TC-SPC-P-002 — Stable dataset: IN CONTROL verdict

Tests: UR-SPC-019, UR-SPC-020
Command: `jrc_spc_p oq/data/p_stable.csv`
Pass criterion: Exit code 0; 'IN CONTROL' or 'STABLE' present in output.

TC-SPC-P-003 — Known p-bar computed correctly

Tests: UR-SPC-019
Command: `jrc_spc_p oq/data/p_stable.csv`

Rationale: With 104 total defectives across 2500 inspected, p_bar = 0.0416.
Pass criterion: Output matches regex 0.04[01234] or 4.[01234]%.

TC-SPC-P-004 — OOC dataset: exit 0, OUT OF CONTROL in output

Tests: UR-SPC-019, UR-SPC-020
Command: `jrc_spc_p oq/data/p_ooc.csv`
Rationale: p_ooc.csv has subgroup 13 = 15 defectives (p = 0.15 > UCL ≈ 0.10).
Pass criterion: Exit code 0; 'OUT OF CONTROL' or 'SIGNALS' present in output.

TC-SPC-P-005 — OOC dataset: subgroup 13 flagged

Tests: UR-SPC-019
Command: `jrc_spc_p oq/data/p_ooc.csv`
Pass criterion: Output contains '13'.

TC-SPC-P-006 — PNG written to ~/Downloads/

Tests: UR-SPC-022
Command: `jrc_spc_p oq/data/p_stable.csv`
Pass criterion: File matching *_jrc_spc_p.png exists in ~/Downloads/ with recent mtime.

TC-SPC-P-007 — No arguments: non-zero exit, usage message

Tests: UR-SPC-026
Command: `jrc_spc_p`
Pass criterion: Exit code ≠ 0; 'Usage' or 'usage' present in output.

TC-SPC-P-008 — File not found: non-zero exit

Tests: UR-SPC-027
Command: `jrc_spc_p /tmp/no_such_file_xyz.csv`
Pass criterion: Exit code ≠ 0; 'not found' or filename present in output.

TC-SPC-P-009 — Missing column: non-zero exit, column named

Tests: UR-SPC-019
Command: `jrc_spc_p oq/data/p_missing_col.csv`
Rationale: p_missing_col.csv lacks the 'defectives' column.
Pass criterion: Exit code ≠ 0; 'defective' present in output.

TC-SPC-P-010 — Defectives exceed n: non-zero exit

Tests: UR-SPC-021
Command: `jrc_spc_p oq/data/p_invalid.csv`
Rationale: p_invalid.csv contains a row where defectives = 101 > n = 100.
Pass criterion: Exit code ≠ 0; one of 'defective', 'exceed', 'greater', 'cannot', or 'invalid' present in output.

TC-SPC-P-011 — Bypass protection: direct Rscript call fails

Tests: UR-SPC-025
Command: `Rscript repos/spc/R/jrc_spc_p.R oq/data/p_stable.csv (RENV_PATHS_ROOT unset)`
Pass criterion: Exit code ≠ 0; 'RENV_PATHS_ROOT' present in output.

6.5 jrc_spc_c

TC-SPC-C-001 — Stable dataset: exit 0, key sections present

Tests: UR-SPC-023, UR-SPC-024
Command: `jrc_spc_c oq/data/c_stable.csv`
Rationale: 25 subgroups; defect counts vary 1–9; no WE rule should fire.
Pass criterion: Exit code 0; output contains c-bar section, UCL section, and verdict.

TC-SPC-C-002 — Stable dataset: IN CONTROL verdict

Tests: UR-SPC-023, UR-SPC-024
Command: `jrc_spc_c oq/data/c_stable.csv`
Pass criterion: Exit code 0; 'IN CONTROL' or 'STABLE' present in output.

TC-SPC-C-003 — Known c-bar computed correctly

Tests: UR-SPC-023
Command: `jrc_spc_c oq/data/c_stable.csv`
Rationale: With 124 total defects across 25 subgroups, c_bar = 4.96.
Pass criterion: Output matches regex 4.8[0-9] or 4.9[0-9].

TC-SPC-C-004 — OOC dataset: exit 0, OUT OF CONTROL in output

Tests: UR-SPC-023, UR-SPC-024
Command: `jrc_spc_c oq/data/c_ooc.csv`
Rationale: c_ooc.csv has subgroup 12 = 15 defects, above UCL ≈ 12.2.
Pass criterion: Exit code 0; 'OUT OF CONTROL' or 'SIGNALS' present in output.

TC-SPC-C-005 — OOC dataset: subgroup 12 flagged

Tests: UR-SPC-023
Command: `jrc_spc_c oq/data/c_ooc.csv`
Pass criterion: Output contains '12'.

TC-SPC-C-006 — PNG written to ~/Downloads/

Tests: UR-SPC-024
Command: `jrc_spc_c oq/data/c_stable.csv`
Pass criterion: File matching *_jrc_spc_c.png exists in ~/Downloads/ with recent mtime.

TC-SPC-C-007 — No arguments: non-zero exit, usage message

Tests: UR-SPC-026
Command: `jrc_spc_c`
Pass criterion: Exit code ≠ 0; 'Usage' or 'usage' present in output.

TC-SPC-C-008 — File not found: non-zero exit

Tests: UR-SPC-027
Command: `jrc_spc_c /tmp/no_such_file_xyz.csv`
Pass criterion: Exit code ≠ 0; 'not found' or filename present in output.

TC-SPC-C-009 — Missing column: non-zero exit, column named

Tests: UR-SPC-023
Command: `jrc_spc_c oq/data/c_missing_col.csv`
Rationale: c_missing_col.csv lacks the 'defects' column.
Pass criterion: Exit code ≠ 0; 'defect' present in output.

TC-SPC-C-010 — Bypass protection: direct Rscript call fails

Tests: UR-SPC-025
Command: `Rscript repos/spc/R/jrc_spc_c.R oq/data/c_stable.csv (RENV_PATHS_ROOT unset)`
Pass criterion: Exit code ≠ 0; 'RENV_PATHS_ROOT' present in output.

7. Acceptance Criteria

The OQ is considered passed when all of the following conditions are met:

- All 55 test cases defined in Section 6 pass with exit code as specified.
- No test case fails (i.e. pytest reports 0 failures and 0 errors).
- The OQ evidence file (timestamped pytest output) is written to ~/.jrscript/<PROJECT_ID>/validation/ by admin_spc_oq.
- All 27 user requirements (UR-SPC-001 through UR-SPC-027) are covered by at least one passing test case.
- Any deviation from this plan is documented and assessed for impact before the OQ is declared passed.

8. Requirements Traceability Matrix

The table below maps each user requirement to the test cases that verify it.

UR	Requirement (summary)	Test Cases
UR-SPC-001	jrc_spc_imr — X-bar, MR-bar, control limits	TC-SPC-IMR-001, 002
UR-SPC-002	jrc_spc_imr — WE rules, violation list	TC-SPC-IMR-002, 003, 004
UR-SPC-003	jrc_spc_imr — --ucl / --lcl	TC-SPC-IMR-005
UR-SPC-004	jrc_spc_imr — IN CONTROL / OUT OF CONTROL verdict	TC-SPC-IMR-001, 002, 003
UR-SPC-005	jrc_spc_imr — PNG to ~/Downloads/	TC-SPC-IMR-006
UR-SPC-006	jrc_spc_imr — reject invalid inputs	TC-SPC-IMR-007, 008, 009, 010
UR-SPC-007	jrc_spc_xbar_r — grand mean, R-bar, limits	TC-SPC-XBR-001, 002
UR-SPC-008	jrc_spc_xbar_r — WE rules, violation list	TC-SPC-XBR-002, 003, 004
UR-SPC-009	jrc_spc_xbar_r — --ucl / --lcl	TC-SPC-XBR-005
UR-SPC-010	jrc_spc_xbar_r — verdict	TC-SPC-XBR-001, 002, 003
UR-SPC-011	jrc_spc_xbar_r — PNG to ~/Downloads/	TC-SPC-XBR-006
UR-SPC-012	jrc_spc_xbar_r — reject unbalanced / n > 10	TC-SPC-XBR-010, 011
UR-SPC-013	jrc_spc_xbar_s — grand mean, S-bar, limits via c4	TC-SPC-XBS-001, 002
UR-SPC-014	jrc_spc_xbar_s — WE rules, violation list	TC-SPC-XBS-002, 003, 004
UR-SPC-015	jrc_spc_xbar_s — --ucl / --lcl	TC-SPC-XBS-005
UR-SPC-016	jrc_spc_xbar_s — verdict	TC-SPC-XBS-001, 002, 003
UR-SPC-017	jrc_spc_xbar_s — PNG to ~/Downloads/	TC-SPC-XBS-006
UR-SPC-018	jrc_spc_xbar_s — reject unbalanced subgroups	TC-SPC-XBS-010
UR-SPC-019	jrc_spc_p — p-bar, variable limits, WE rules	TC-SPC-P-001, 002, 003, 004, 005
UR-SPC-020	jrc_spc_p — verdict	TC-SPC-P-001, 002, 004
UR-SPC-021	jrc_spc_p — reject defectives > n	TC-SPC-P-010
UR-SPC-022	jrc_spc_p — PNG to ~/Downloads/	TC-SPC-P-006
UR-SPC-023	jrc_spc_c — c-bar, sigma, limits, WE rules	TC-SPC-C-001, 002, 003, 004, 005
UR-SPC-024	jrc_spc_c — verdict; PNG to ~/Downloads/	TC-SPC-C-001, 002, 004, 006
UR-SPC-025	All scripts — RENV_PATHS_ROOT bypass protection	TC-SPC-IMR-011, XBR-012, XBS-011, P-011, C-010
UR-SPC-026	All scripts — no-arguments usage message	TC-SPC-IMR-007, XBR-007, XBS-007, P-007, C-007
UR-SPC-027	All scripts — file not found error	TC-SPC-IMR-008, XBR-008, XBS-008, P-008, C-008

9. Roles and Responsibilities

Role	Responsibility
Author	Prepares this validation plan; executes the OQ test suite; documents results in the validation report.
Reviewer	Reviews this plan and the corresponding validation report for technical accuracy and completeness.
Approver (QA)	Approves this plan prior to OQ execution; approves the validation report upon successful completion of OQ.

10. Deviation Handling

Any deviation from this plan (e.g. a test case that cannot be executed as written, or a test that fails due to an environmental issue rather than a software defect) must be documented in the validation report with:

- A unique deviation ID
- Description of the deviation
- Root cause assessment
- Impact on the OQ conclusion
- Disposition (accepted with justification, or requires remediation)

The OQ may not be declared passed if any test case fails due to a software defect, unless a formal deviation with justification is approved by the QA approver.